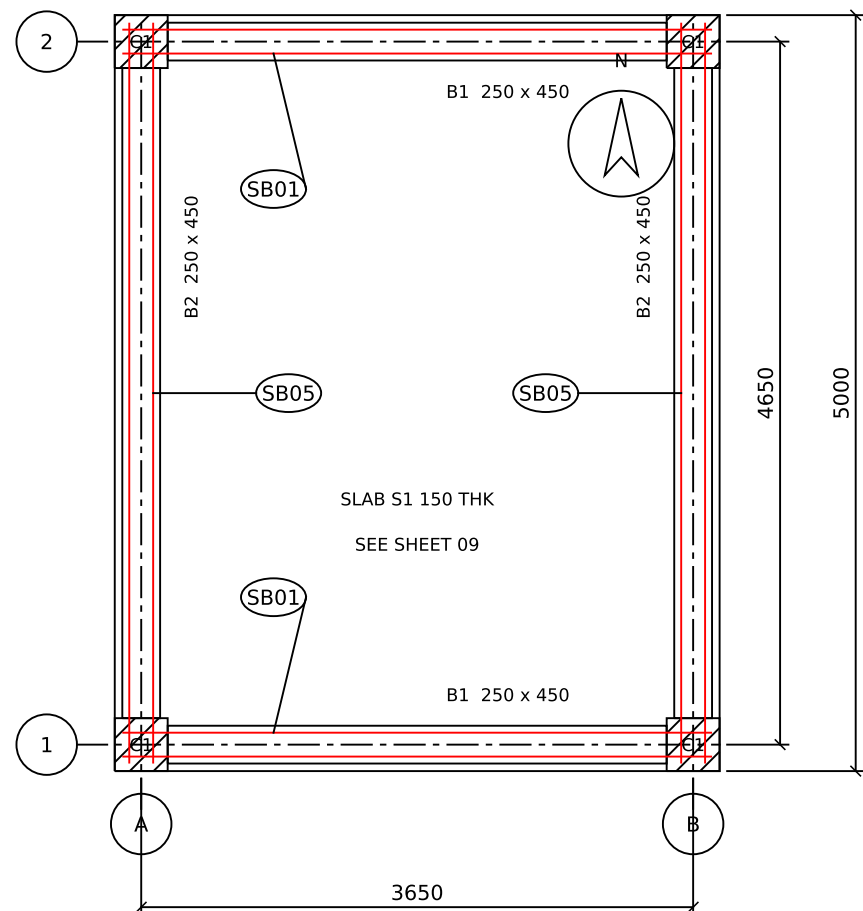
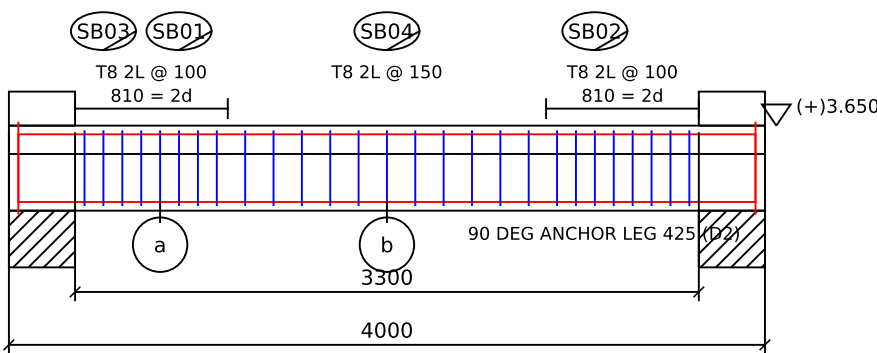




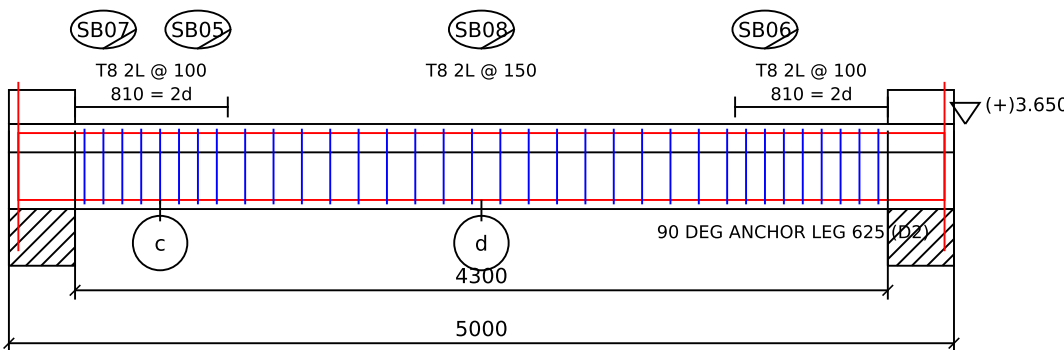
1 SENTRY POST - FIRST-FLOOR FRAMING PLAN

1 : 50 EXTERNAL 4000 x 5000, GRIDS A-B 3650 c/c, 1-2 4650 c/c



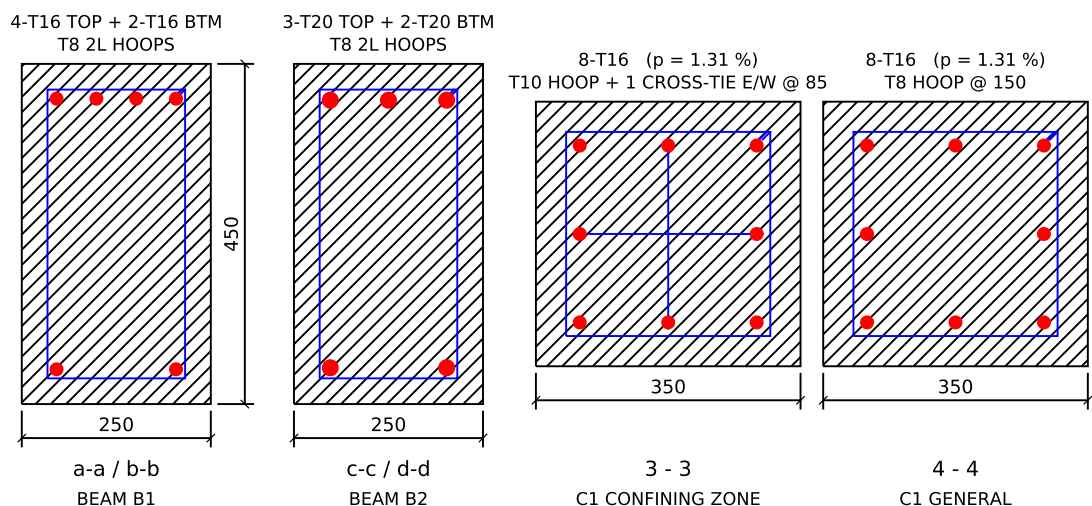
2 BEAM B1 - LONGITUDINAL SECTION (GRIDS 1 AND 2)

1 : 40 B1 250 x 450, CLEAR SPAN 3300, d = 404, 4-T16 TOP + 2-T16 BOTTOM (D1)



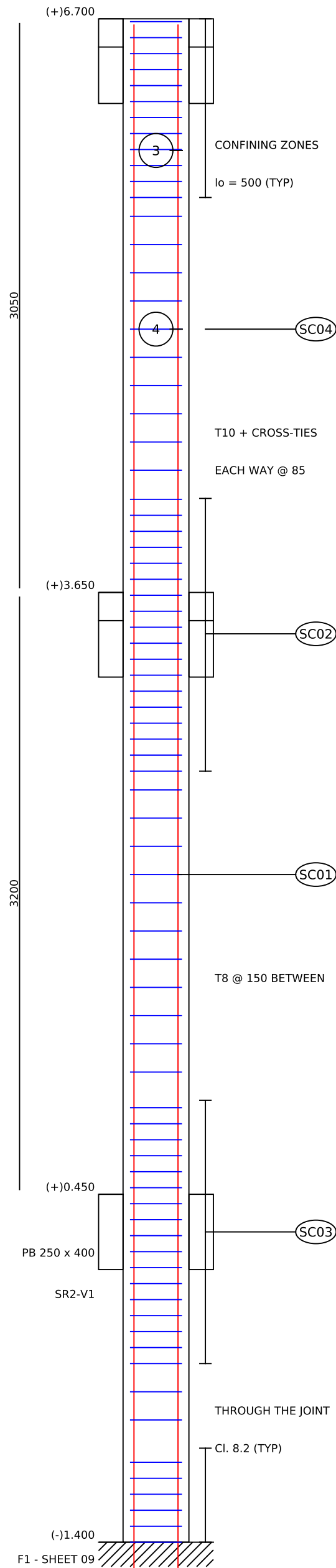
3 BEAM B2 - LONGITUDINAL SECTION (GRIDS A AND B)

1 : 40 B2 250 x 450, CLEAR SPAN 4300, d = 404, 3-T20 TOP + 2-T20 BOTTOM (D1)



4 SECTIONS - BEAMS AND COLUMN

1 : 10 COVER 30 TO BEAMS, 40 TO COLUMNS



5 COLUMN C1

1 : 25 VERTICAL SECTION, FULL HEIGHT

BEAM SCHEDULE - SENTRY POST FRAME								
BEAM	SIZE	SPAN c/c	CLEAR	COVER	d	TOP AT SUPP.	BOTTOM	HOOPS
B1 GRIDS 1 AND 2	250 x 450	3650	3300	30	404	4-T16	2-T16	T8 2L @ 100 / 150
B2 GRIDS A AND B	250 x 450	4650	4300	30	404	3-T20	2-T20	T8 2L @ 100 / 150
IS 13920 Cl. 6.1: b 250 >= 200 OK b/D 0.556 >= 0.3 OK Ln/D 7.33 >= 4 OK								
SCHEDULED HERE, DETAILED ELSEWHERE, NOT DRAWN ON THIS SHEET								
PB PLINTH BEAM	250 x 400	3650 / 4650	-	30	354	3-T12	3-T12	NOT STATED - SR2-V1

COLUMN SCHEDULE							
COL	SIZE	LEVELS	COVER	VERTICALS	p	CONFINING (Cl. 8.1)	GENERAL TIES
C1	350 x 350	(-)1.400 TO +6.700	40	8-T16 (1608)	1.31 %	T10 + 1 CROSS-TIE E/W @ 85	T8 @ 150
4 No. COLUMNS. SHORT COLUMN: lex/D = 9.43 < 12 (IS 456 Cl. 25.1.2)							
CONFINING STEEL OVER lo = 500 FROM EVERY JOINT FACE AND THROUGH THE JOINT							
Ash PROVIDED 3 x 78.5 = 235.5 mm2 vs 168.5 REQUIRED (IS 13920 Cl. 8.1)							

BAR MARK SCHEDULE - BEAMS B1 / B2 AND COLUMN C1					
MARK	BAR	SPACING	LOCATION	CUT mm	No.
SB01	T16	-	TOP, 4 BARS, FULL LENGTH (D1)	4750	8
SB02	T16	-	BOTTOM, 2 BARS CONTINUOUS	4750	4
SB03	T8	100	HOOPS, HINGE ZONE 810 = 2d EACH END	1320	32
SB04	T8	150	HOOPS, CENTRAL ZONE	1320	22
SB05	T20	-	TOP, 3 BARS, FULL LENGTH (D1)	6150	6
SB06	T20	-	BOTTOM, 2 BARS CONTINUOUS	6150	4
SB07	T8	100	HOOPS, HINGE ZONE 810 = 2d EACH END	1320	32
SB08	T8	150	HOOPS, CENTRAL ZONE	1320	36
SC01	T16	-	VERTICALS, 8 PER COLUMN, NO SPLICE (D3)	8724	32
SC02	T10	85	CONFINING HOOPS, 9 ZONES (IS 13920 Cl. 8.1)	1280	212
SC03	T10	85	CROSS-TIES, ONE EACH WAY WITH EVERY HOOP	470	424
SC04	T8	150	GENERAL TIES, OUTSIDE THE CONFINING ZONES	1240	92

IS 13920:2016 COMPLIANCE - SENTRY POST FRAME			
CLAUSE	REQUIREMENT	PROVIDED	VERDICT
Cl. 6.1.1 / 6.1.2 / 6.1.3	b >= 200, b/D >= 0.3, Ln/D >= 4	250, 0.556, 7.33	PASS
Cl. 6.2.1(b) / 6.2.2	rho,min 266 mm2 ; rho,max 2525 mm2	402 (2-T16) ; 942 (3-T20) MAX	PASS
Cl. 6.2.3 positive at joint	>= 50 % of the negative steel	402 = 50 % of 804	PASS
Cl. 6.2.4 through the joint	>= 25 % of the max = 201 mm2	402 CONTINUOUS	PASS
Cl. 6.3.3 capacity shear	Vu B1 120.4 / B2 144.7 kN	T8 2-LEG @ 100	PASS
Cl. 6.3.4 tau_c = 0	EQ share B1 60 % / B2 49.7 %	TAKEN AS TRIGGERED	PASS
Cl. 6.3.5.1 / 6.3.5.2	hinge min(d/4=101, 8phi=128) ; else d/2=202	100 OVER 2d = 810 ; 150	PASS
Cl. 7.2.1 SCWB, FIRST FLOOR	1.4 sum Mb 171.6 (X) / 195.7 (Z)	sum Mc 213.9	PASS
Cl. 7.2.1 SCWB, ROOF	1.4 sum Mb 138.5 AT 2-T20	sum Mc 101	MARGINAL - SR2-F1
Cl. 7.4 ties, full length	<= 0.5 x least dimension = 175	150	PASS
Cl. 8.1 lo and spacing	lo max(350, h/6=458, 450) ; 75 <= s <= 87.5	500 ; 85	PASS
Cl. 8.1 Ash	168.5 mm2 (minimum 68.9)	235.5 (3 LEGS T10)	PASS
Cl. 8.2 through the joint	CONFINING STEEL CONTINUED	PROVIDED	PASS
Cl. 10.4 boundary elements	CHECKED FOR THE BOX ONLY	NOT TRIGGERED	N/A HERE

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA
STRUCTURAL - PHASE 2 + M1

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS.
- WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH AND STR DRGS.
- SENTRY POST CONCRETE M30. REINFORCEMENT Fe500D TO IS 1786:2008. COVER 40 TO COLUMNS, 30 TO BEAMS AND SLABS, 50 TO FOOTINGS (IS 456 Cl. 26.4.2 / TABLE 16).
- THE SENTRY POST FRAME IS THE IS 13920:2016 ELEMENT OF THIS PROJECT. IT IS A SPECIAL MOMENT-RESISTING FRAME, R = 3.0, Ah = 0.100, DESIGNED ON THE STAAD BASE SHEAR Vb = 73.18 kN.
- DUCTILE DETAILING GOVERNS EVERY BAR ON THIS SHEET. Cl. 6.2.3 SETS THE BOTTOM STEEL. Cl. 6.3.3 / 6.3.4 SET THE HOOPS (tau_c TAKEN AS ZERO IN THE HINGE REGIONS), Cl. 6.3.5.1 SETS 100 c/c OVER 2d = 810, AND Cl. 8.1 / 8.2 SET THE COLUMN CONFINING STEEL.
- THE TAGS (D1), (D2) AND (D3) ON THIS SHEET ARE DECLARED DETAILING DECISIONS, AND EVERY 'SR2-' REFERENCE IS AN OPEN ITEM. BOTH ARE RECORDED IN MASTER PART H.40.4 AND H.40.5. D1: TOP STEEL IS DETAILED CONTINUOUS OVER THE FULL SPAN BECAUSE THE MASTER GIVES 'TOP AT SUPPORTS' AND NO CURTAILMENT POINT.
- THIS SHEET DETAILS THE FIRST-FLOOR FRAME ONLY. THE ROOF FRAME IS NOT DETAILED - SEE FINDING SR2-F1 IN MASTER PART H.40.5.
- THE SENTRY POST IS NOT DESIGNED FOR THE BLAST, AND THAT IS A RECORDED DECISION (MASTER B.8). REFLECTED FORCE ON THE 4.0 x 6.7 m FACE IS 36 600 kN; HARDENING WOULD NEED ABOUT 700 mm OF RC ON ALL FOUR ABOVE-GROUND FACES.
- CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

STRUCTURAL REINFORCEMENT DETAILING OF SENTRY POST - BEAMS & COLUMN

CBRN HARDENED UG OPS ROOM

SHEET 08

Project number 01

Date 16 SEP 2026

Drawn by SYN 01

Checked by DR IR CHAUDHARI

STR008

Scale As indicated